

torque can be provided is shown in the Fig. 1 in which the switch S after disconnecting from the supply is thrown to terminals 1'2'. The machine then gets connected to a non-inductive load resistance R_L . The power drawn by this resistance will acts as a retarding torque on the armature which will make it slow more quickly.

The additional loss in the resistance will be equal to product of ammeter reading and the average reading of the voltmeter (for a fall of 5% of voltmeter reading, the time is noted.) The ammeter reading is also changing so its average reading is taken. Thus the additional losses is $I_a^2 (R_a + R)$. Let t_1 be the time when armature is considered alone and t_2 be the time when armature is connected across a load resistance, V be average voltage across R and I_a be the average current and W' is additional retarding electrical torque supplied by motor.

$$W = \left(\frac{2\pi}{60} \right)^2 I N \frac{dN}{dt_1}$$

$$W + W' = \left(\frac{2\pi}{60} \right)^2 I N \frac{dN}{dt_2}$$

If dN i.e. change in speed is same in two cases then

$$\frac{W + W'}{W} = \frac{1/dt_2}{1/dt_1} = \frac{dt_1}{dt_2} = \frac{t_1}{t_2}$$

$$\therefore (W + W') t_2 = W t_1$$

$$\therefore W (t_1 - t_2) = W' t_2$$

$$\therefore \boxed{W = W' \frac{t_2}{t_1 - t_2}}$$

Here dN/dt_1 is rate of change in speed without extra load whereas dN/dt_2 is rate change in speed with extra electrical load which provides retarding torque.

UNIT IV

SINGLE PHASE TRANSFORMERS

WORKING PRINCIPLE OF A TRANSFORMER:

The basic principle on which the transformer works is **Faraday's Law of Electromagnetic Induction** or mutual induction between the two coils. The working of the transformer is explained below. The transformer consists of two separate windings placed over the laminated silicon steel core.

The winding to which AC supply is connected is called primary winding and to which load is connected is called secondary winding as shown in the figure below. It works on the alternating current only because an alternating flux is required for mutual induction between the two windings.

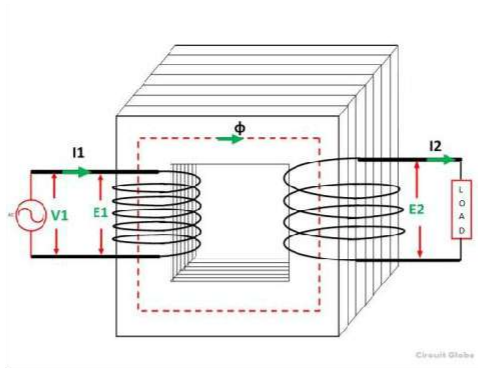


Fig: Simple Transformer

When the AC supply is given to the primary winding with a voltage of V_1 , an alternating flux ϕ sets up in the core of the transformer, which links with the secondary winding and as a result of it, an emf is induced in it called Mutually Induced emf. The direction of this induced emf is opposite to the applied voltage V_1 , this is because of the Lenz's law shown in the figure below

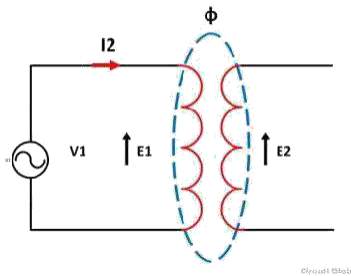


Fig: Transformer Symbol

Physically, there is no electrical connection between the two windings, but they are magnetically connected. Therefore, the electrical power is transferred from the primary circuit to the secondary circuit through mutual inductance. The induced emf in the primary and secondary windings depends upon the rate of change of flux linkage that is $(N \frac{d\phi}{dt})$.

$\frac{d\phi}{dt}$ is the change of flux and is same for both the primary and secondary windings. The induced emf E_1 in the primary winding is proportional to the number of turns N_1 of the primary windings ($E_1 \propto N_1$). Similarly induced emf in the secondary winding is proportional to the number of turns on the secondary side. ($E_2 \propto N_2$).

Transformer on DC supply:

As discussed above, the transformer works on AC supply, and it cannot work on DC supply. If the rated DC voltage is applied across the primary winding, a constant magnitude flux will set up in the core of the transformer and hence there will not be any self-induced emf generation, as for the linkage of flux with the secondary winding there must be an alternating flux required and not a constant flux.

According to Ohm's Law

$$\text{Primary Current} = \frac{\text{DC applied voltage}}{\text{Resistance of primary winding}}$$

The resistance of the primary winding is very low, and the primary current is high. So this current is much higher than the rated full load primary winding current. Hence, as a result, the amount of heat produced will be greater and therefore, eddy current loss (I^2R) loss will be more. Because of this, the insulations of the primary windings will get burnt, and the transformer will be damaged.

Turn Ratio:

It is defined as the ratio of primary to secondary turns.

$$\text{Turn ratio} = \frac{N_1}{N_2}$$

If $N_2 > N_1$ the transformer is called Step up transformer

If $N_2 < N_1$ the transformer is called Step down transformer

Transformation Ratio:

The transformation ratio is defined as the ratio of the secondary voltage to the primary voltage. It is denoted by K.

$$K = \frac{E_2}{E_1} = \frac{N_2}{N_1}$$

As ($E_2 \propto N_2$ and $E_1 \propto N_1$)

Ideal Transformer:

Definition: The transformer which is free from all types of losses is known as an ideal transformer. It is an imaginary transformer which has no core loss, no ohmic resistance and no leakage flux. The ideal transformer has the following important characteristic.

1. The resistance of their primary and secondary winding becomes zero.
2. The core of the ideal transformer has infinite permeability. The infinite permeable means less magnetising current requires for magnetising their core.
3. The leakage flux of the transformer becomes zero, i.e. the whole of the flux induces in the core of the transformer links with their primary and secondary winding.
4. The ideal transformer has 100 percent efficiency, i.e., the transformer is free from hysteresis and eddy current loss.

The above mention properties are not possible in the practical transformer. In an ideal transformer, there is no power loss. Therefore, the output power is equal to the input power.

$$E_2 I_2 \cos\phi = E_1 I_1 \cos\phi \quad \text{or} \quad E_2 I_2 = E_1 I_1$$

OR

$$\frac{E_2}{E_1} = \frac{I_1}{I_2}$$

Since $E_1 \propto N_2$ and $E_1 \propto N_1$, also E_1 is similar to V_1 and E_2 is similar to V_2

Therefore, transformation ratio will be given by the equation shown below

$$\frac{V_2}{V_1} = \frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{I_1}{I_2} = K$$

The primary and the secondary currents are inversely proportional to their respective turns.

Behaviour of Ideal Transformer:

Consider the ideal transformer shown in the figure below. The voltage source V_1 is applied across the primary winding of the transformer. Their secondary winding is kept open. The N_1 and N_2 are the numbers of turns of their primary and secondary winding.

The current I_m is the magnetizing current flows through the primary winding of the transformer. The magnetizing current produces the flux ϕ_m in the core of the transformer. As the permeability of the core is infinite, the flux of the core link with both the primary and secondary winding of the transformer.

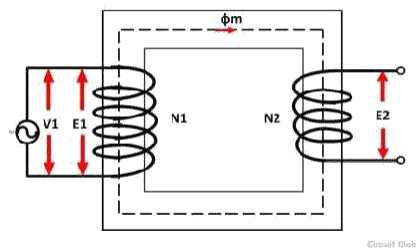
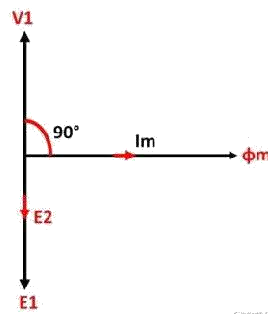


Fig: Ideal Trasformer

The flux link with the primary winding induces the emf E_1 because of self-induction. The direction of the induced emf is inversely proportional to the applied voltage V_1 . The emf E_2 induces in the secondary winding of the transformer because of mutual induction.\

Phasor Diagram of Ideal Transformer:

The phasor diagram of the ideal transformer is shown in the figure below. As the coil of the primary transformer is purely inductive the magnetising current induces in the transformer lag 90° by the input voltage V_1 . The E_1 and E_2 are the emf induced in the primary and secondary winding of the transformer. The direction of the induces emf inversely proportional to the applied voltage



Phasor Diagram of an Ideal Transformer

Point to Remember

The input energy of the transformer is equal to their output energy. The power loss in the ideal transformer becomes zero.

EMF EQUATION OF A TRANSFORMER:

When a sinusoidal voltage is applied to the primary winding of a transformer, alternating flux ϕ_m sets up in the iron core of the transformer. This sinusoidal flux links with both primary and secondary winding.

The function of flux is a sine function. The rate of change of flux with respect to time is derived mathematically.

The derivation of **EMF Equation** of the transformer is shown below. Let

- ϕ_m be the maximum value of flux in Weber
- f be the supply frequency in Hz
- N_1 is the number of turns in the primary winding
- N_2 is the number of turns in the secondary winding

Φ is the flux per turn in Weber

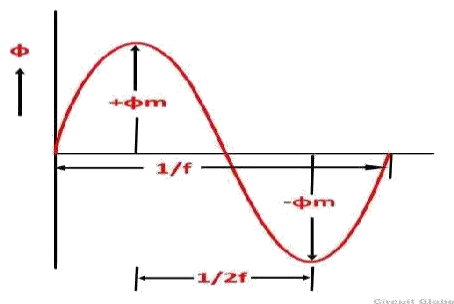


Fig: Flux waveform

As shown in the above figure that the flux changes from $+\phi_m$ to $-\phi_m$ in half a cycle of $1/2f$ seconds. By Faraday's Law

Let E_1 is the emf induced in the primary winding

$$E_1 = -\frac{d\psi}{dt} \dots \dots \dots (1)$$

Where $\Psi = N_1\phi$

$$\text{Therefore, } E_1 = -N_1 \frac{d\phi}{dt} \dots \dots \dots (2)$$

Since ϕ is due to AC supply $\phi = \phi_m \sin \omega t$

$$E_1 = -N_1 \frac{d}{dt} (\phi_m \sin \omega t)$$

$$E_1 = -N_1 \omega \phi_m \cos \omega t$$

$$E_1 = N_1 \omega \phi_m \sin(\omega t - \pi/2) \dots \dots \dots (3)$$

So the induced emf lags flux by 90 degrees.

Maximum value of emf

$$E_{1\max} = N_1 \omega \phi_m \dots \dots \dots (4)$$

But $\omega = 2\pi f$

$$E_{1\max} = 2\pi f N_1 \phi_m \dots \dots \dots (5)$$

Root mean square RMS value is

$$E_1 = \frac{E_{1\max}}{\sqrt{2}} \dots \dots \dots (6)$$

Putting the value of $E_{1\max}$ in equation (6) we get

$$E_1 = \sqrt{2} \pi f N_1 \phi_m \dots \dots \dots (7)$$

Putting the value of $\pi = 3.14$ in the equation (7) we will get the value of E_1 as

$$E_1 = 4.44fN_1\phi_m \dots \dots (8)$$

Similarly

$$E_2 = \sqrt{2}\pi fN_2\phi_m$$

Or

$$E_2 = 4.44fN_2\phi_m \dots \dots (9)$$

Now, equating the equation (8) and (9) we get

$$\frac{E_2}{E_1} = \frac{4.44fN_2\phi_m}{4.44fN_1\phi_m}$$

Or

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

The above equation is called the turn ratio where K is known as transformation ratio.

The equation (8) and (9) can also be written as shown below using the relation

($\phi_m = B_m \times A_i$) where A_i is the iron area and B_m is the maximum value of flux density.

$$E_1 = 4.44N_1fB_mA_i \text{ Volts} \quad \text{and} \quad E_2 = 4.44N_2fB_mA_i \text{ Volts}$$

For a sinusoidal wave

$$\frac{\text{R. M. S value}}{\text{Average value}} = \text{Form factor} = 1.11$$

Magnetic Leakage flux:

In a transformer it is observed that, all the flux linked with primary winding does not get linked with secondary winding. A small part of the flux completes its path through air rather than through the core (as shown in the fig at right), and this small part of flux is called as leakage flux or magnetic leakage in transformers. This leakage flux does not link with both the windings, and hence it does not contribute to transfer of energy from primary winding to secondary winding. But, it produces self induced emf in each winding. Hence, leakage flux produces an effect equivalent to an inductive coil in series with each winding. And due to this there will be leakage reactance.

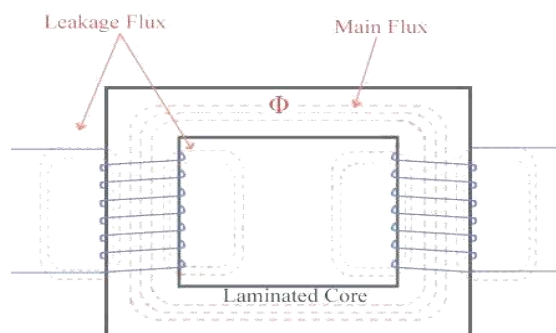


Fig: Magnetic Leakage flux

(To minimize this leakage reactance, primary and secondary windings are not placed on separate legs, refer the diagram of core type and shell type transformer from construction of transformer.)

Practical Transformer with Resistance and Leakage Reactance

In the following figure, leakage reactance and resistance of the primary winding as well as secondary winding are taken out, representing a practical transformer.

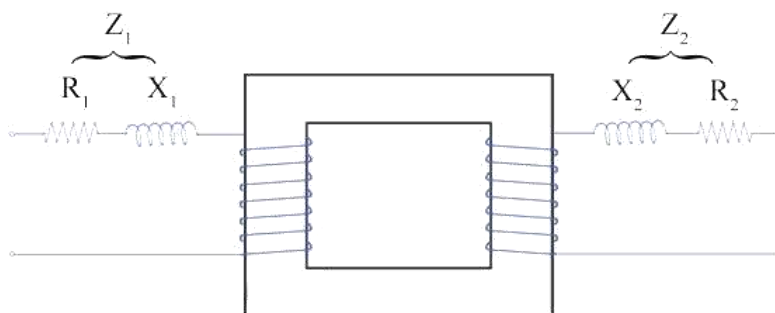


Fig: Practical Transformer with Resistance And Leakage Reactance

Where, R_1 and R_2 = resistance of primary and secondary winding respectively

X_1 and X_2 = **leakage reactance** of primary and secondary winding resp.

Z_1 and Z_2 = Primary impedance and secondary impedance resp.

$$Z_1 = R_1 + jX_1 \text{ ...and } Z_2 = R_2 + jX_2 .$$

The impedance in each winding lead to some voltage drop in each winding. Considering this voltage drop the **voltage equation of transformer** can be given as –

$$V_1 = E_1 + I_1(R_1 + jX_1) \text{ -----primary side}$$

$$V_2 = E_2 - I_2(R_2 + jX_2) \text{ -----secondary side}$$

where, V_1 = supply voltage of primary winding

V_2 = terminal voltage of secondary winding

E_1 and E_2 = induced emf in primary and secondary winding respectively.

Resistance and Reactance of the Transformer:

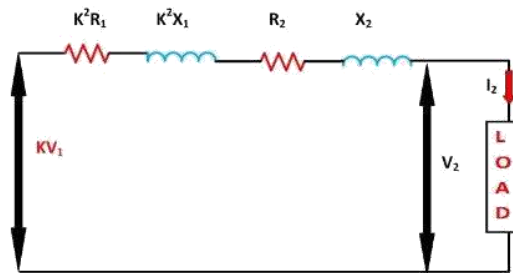
The Resistance of the transformer is defined as the internal resistance of both primary and secondary windings. In an actual transformer, the primary and the secondary windings have some resistance represented by R_1 and R_2 and the reactances by X_1 and X_2 . Let K be the transformation ratio. To make the calculations easy the resistances and reactances can be transferred to either side that means either all the primary terms are referred to the secondary side, or all the secondary terms are referred to the primary side.

The resistive and the reactive drops in the primary and secondary side are represented as follows

- Resistive drop in the secondary side = $I_2 R_2$
- Reactive drop in the secondary side = $I_2 X_2$
- Resistive drop in the primary side = $I_1 R_1$
- Reactive drop in the primary side = $I_1 X_1$

Primary Side Referred to Secondary Side

Since the transformation ratio is K, primary resistive and reactive drop as referred to secondary side will be K times, i.e., $K I_1 R_1$ and $K I_1 X_1$ respectively. If I_1 is substituted equal to $K I_2$ then we have primary resistive and reactive drop referred to secondary side equal to $K^2 I_2 R_1$ and $K^2 I_2 X_1$ respectively.



The Total resistive drop in a transformer

$$K^2 I_2 R_1 + I_2 R_2 = I_2 (K^2 R_1 + R_2)$$

Total reactive drop in a transformer

$$K^2 I_2 X_1 + I_2 X_2 = I_2 (K^2 X_1 + X_2)$$

The term

$$(K^2 R_1 + R_2) \text{ And } (K^2 X_1 + X_2)$$

represent the equivalent resistance and reactance of the transformer referred to the secondary side.

TRANSFORMER ON NO-LOAD CONDITION:

When the transformer is operating at no load, the secondary winding is open circuited, which means there is no load on the secondary side of the transformer and, therefore, current in the secondary will be zero, while primary winding carries a small current I_0 called no load current which is 2 to 10% of the rated current. This current is responsible for supplying the iron losses (hysteresis and eddy current losses) in the core and a very small amount of copper losses in the primary winding. The angle of lag depends upon the losses in the transformer. The power factor is very low and varies from 0.1 to 0.15.

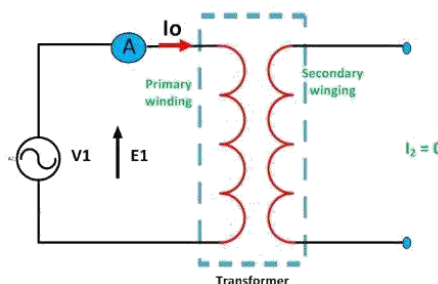


Fig: Transformer is operating at no load

The no load current consists of two components

- Reactive or magnetizing component I_m
(It is in quadrature with the applied voltage V_1 . It produces flux in the core and does not consume any power)
- Active or power component I_w , also known as working component
(It is in phase with the applied voltage V_1 . It supplies the iron losses and a small amount of primary copper loss)

The following steps are given below to draw the phasor diagram

1. The function of the magnetizing component is to produce the magnetizing flux, and thus, it will be in phase with the flux.
2. Induced emf in the primary and the secondary winding lags the flux ϕ by 90 degrees.
3. The primary copper loss is neglected, and secondary current losses are zero as $I_2 = 0$. Therefore, the current I_0 lags behind the voltage vector V_1 by an angle ϕ_0 called no-load power factor angle shown in the phasor diagram above.
4. The applied voltage V_1 is drawn equal and opposite to the induced emf E_1 because the difference between the two, at no load, is negligible.
5. Active component I_w is drawn in phase with the applied voltage V_1 .
6. The phasor sum of magnetizing current I_m and the working current I_w gives the no load current I_0 .

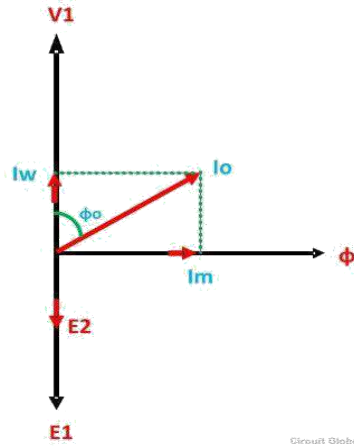


Fig: Phasor diagram drawn

Working component $I_w = I_0 \cos \phi_0$

No load current $I_0 = \sqrt{I_w^2 + I_m^2}$

Magnetizing component $I_m = I_0 \sin \phi_0$

Power factor $\cos \phi_0 = \frac{I_w}{I_0}$

No load power input $P_0 = V_1 I_0 \cos \phi_0$

TRANSFORMER ON LOAD CONDITION:

When the transformer is on loaded condition, the secondary of the transformer is connected to load. The load can be resistive, inductive or capacitive. The current I_2 flows through the secondary winding of the transformer. The magnitude of the secondary current depends on the terminal voltage V_2 and the load impedance. The phase angle between the secondary current and voltage depends on the nature of the load.

Operation of the Transformer on Load Condition

The Operation of the Transformer on Load Condition is explained below

- When secondary of the transformer is kept open, it draws the no-load current from the main supply. The no-load current induces the magneto motive force $N_0 I_0$ and this force set up the flux Φ in the core of the transformer. The circuit of the transformer at no load condition is shown in the figure below.

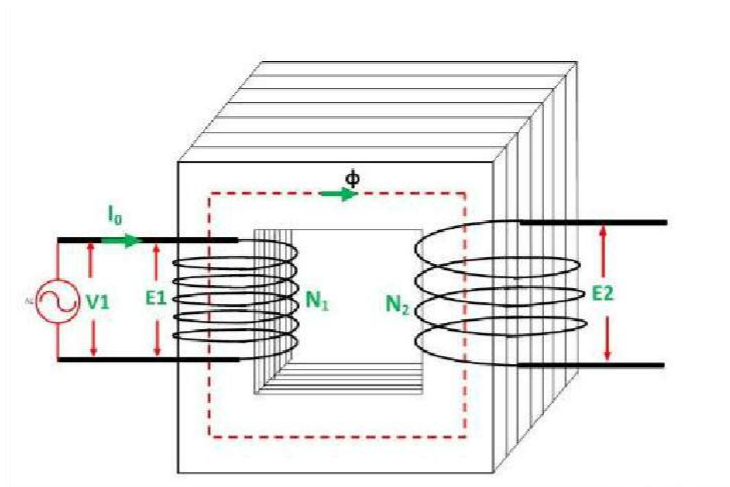


Fig: Operation of the Transformer on Load Condition

- When the load is connected to the secondary of the transformer, the I_2 current flows through their secondary winding. The secondary current induces the magnetomotive force $N_2 I_2$ on the secondary winding of the transformer. This force set up the flux ϕ_2 in the transformer core. The flux ϕ_2 oppose the flux ϕ , according to Lenz's law

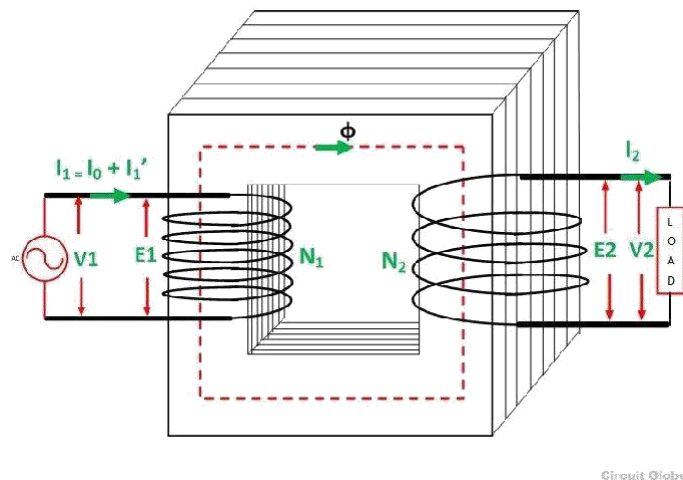


Fig: Operation of the Transformer on Load Condition

- As the flux ϕ_2 opposes the flux ϕ , the resultant flux of the transformer decreases and this flux reduces the induced EMF E_1 . Thus, the strength of the V_1 is more than E_1 and an additional primary current I'_1 drawn from the main supply. The additional current is used for restoring the original value of the flux in the core of the transformer so that the $V_1 = E_1$. The primary current I'_1 is in phase opposition with the secondary current I_2 . Thus, it is called the primary counter balancing current.
- The additional current I'_1 induces the magnetomotive force $N_1 I'_1$. And this force set up the flux ϕ'_1 . The direction of the flux is same as that of the ϕ and it cancels the flux ϕ_2 which induces because of the MMF $N_2 I_2$

Now, $N_1 I_1' = N_2 I_2$

$$I_1' = \left(\frac{N_2}{N_1} \right) I_2 = K I_2$$

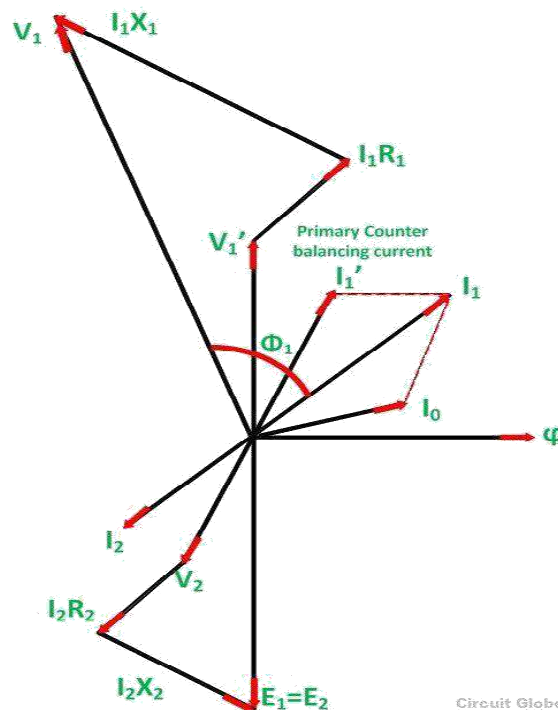
Therefore,

- The phasor difference between V_1 and I_1 gives the power factor angle ϕ_1 of the primary side of the transformer.
- The power factor of the secondary side depends upon the type of load connected to the transformer.
- If the load is inductive as shown in the above phasor diagram, the power factor will be lagging, and if the load is capacitive, the power factor will be leading. The total primary current I_1 is the vector sum of the current I_0 and I_1' . i.e

$$\bar{I}_1 = \bar{I}_0 + \bar{I}_1'$$

Phasor Diagram of Transformer on Inductive Load:

The phasor diagram of the actual transformer when it is loaded inductively is shown below



Phasor Diagram of the Transformer on Inductive Load

Steps to draw the phasor diagram

- Take flux ϕ a reference
- Induces emf E_1 and E_2 lags the flux by 90 degrees.
- The component of the applied voltage to the primary equal and opposite to induced emf in the primary winding. E_1 is represented by V_1' .
- Current I_0 lags the voltage V_1' by 90 degrees.
- The power factor of the load is lagging. Therefore current I_2 is drawn lagging E_2 by an angle ϕ_2 .

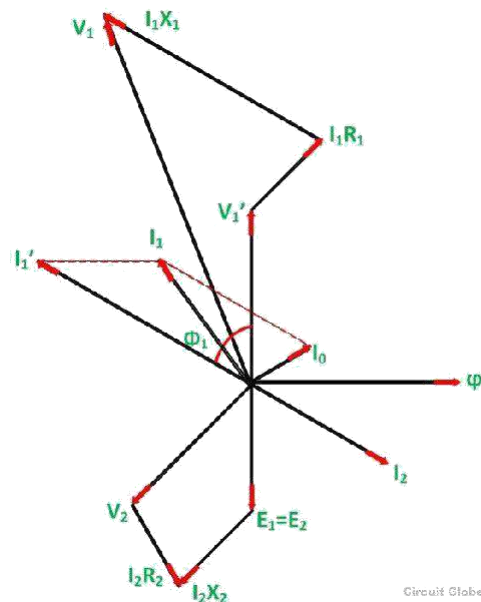
- The resistance and the leakage reactance of the windings result in a voltage drop, and hence secondary terminal voltage V_2 is the phasor difference of E_2 and voltage drop.
 $V_2 = E_2 - \text{voltage drops}$
 $I_2 R_2$ is in phase with I_2 and $I_2 X_2$ is in quadrature with I_2 .
- The total current flowing in the primary winding is the phasor sum of I_1' and I_0 .
- Primary applied voltage V_1 is the phasor sum of V_1' and the voltage drop in the primary winding.
- Current I_1' is drawn equal and opposite to the current
 $I_2 V_1 = V_1' + \text{voltage drop}$
 $I_1 R_1$ is in phase with I_1 and $I_1 X_1$ is in quadrature with I_1 .
- The phasor difference between V_1 and I_1 gives the power factor angle ϕ_1 of the primary side of the transformer.
- The power factor of the secondary side depends upon the type of load connected to the transformer.
- If the load is inductive as shown in the above phasor diagram, the power factor will be lagging, and if the

load is capacitive, the power factor will be leading. Where $I_1 R_1$ is the resistive drop in the primary windings

$I_2 X_2$ is the reactive drop in the secondary winding

Phasor Diagram of Transformer on Capacitive Load

The Transformer on Capacitive load (leading power factor load) is shown below in the phasor diagram.



Phasor Diagram of the Transformer on Capacitive Load

Steps to draw the phasor diagram at capacitive load

- Take flux ϕ a reference
- Induces emf E_1 and E_2 lags the flux by 90 degrees.
- The component of the applied voltage to the primary equal and opposite to induced emf in the primary winding. E_1 is represented by V_1' .
- Current I_0 lags the voltage V_1' by 90 degrees.

- The power factor of the load is leading. Therefore current I_2 is drawn leading E_2
- The resistance and the leakage reactance of the windings result in a voltage drop, and hence secondary terminal voltage V_2 is the phasor difference of E_2 and voltage drop.

$$V_2 = E_2 - \text{voltage drops}$$

$I_2 R_2$ is in phase with I_2 and $I_2 X_2$ is in quadrature with I_2 .

- Current I_1' is drawn equal and opposite to the current I_2
- The total current I_1 flowing in the primary winding is the phasor sum of I_1' and I_0 .
- Primary applied voltage V_1 is the phasor sum of V_1' and the voltage drop in the primary winding.

$$V_1 = V_1' + \text{voltage drop}$$

$I_1 R_1$ is in phase with I_1 and $I_1 X_1$ is in quadrature with I_1 .

- The phasor difference between V_1 and I_1 gives the power factor angle ϕ_1 of the primary side of the transformer.
- The power factor of the secondary side depends upon the type of load connected to the transformer.

EQUIVALENT CIRCUIT OF A TRANSFORMER:

The equivalent circuit diagram of any device can be quite helpful in predetermination of the behaviour of the device under the various condition of operation. It is simply the circuit representation of the equation describing the performance of the device.

The simplified equivalent circuit of a transformer is drawn by representing all the parameters of the transformer either on the secondary side or on the primary side. The equivalent circuit diagram of the transformer is shown below

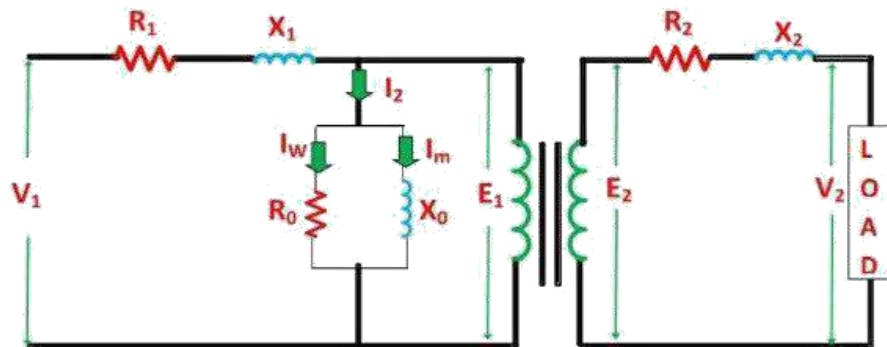


Fig: Equivalent circuit diagram of a transformer

Let the equivalent circuit of a transformer having the transformation ratio $K = E_2/E_1$

The induced emf E_1 is equal to the primary applied voltage V_1 less primary voltage drop. This voltage causes current I_0 no load current in the primary winding of the transformer. The value of no-load current is very small, and thus, it is neglected. Hence, $I_1 = I_1'$. The no load current is further divided into two components called magnetizing current (I_m) and working current (I_w).

These two components of no-load current are due to the current drawn by a noninductive resistance R_0 and pure reactance X_0 having voltage E_1 or (V_1 – primary voltage drop).

The secondary current I_2 is

$$I_2 = \frac{I_1'}{K} = \frac{I_1 - I_0}{K}$$

The terminal voltage V_2 across the load is equal to the induced emf E_2 in the secondary winding less voltage drop in the secondary winding.

Equivalent Circuit when all the Quantities are referred to Primary side:

In this case to draw the equivalent circuit of the transformer all the quantities are to be referred to the primary as shown in the figure below

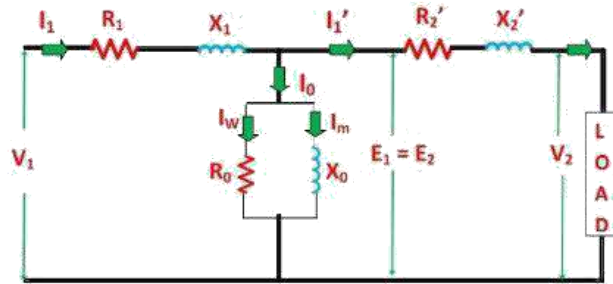


Fig: Circuit Diagram of Transformer when all the Secondary Quantities are Referred to Primary Side

The following are the values of resistance and reactance given below
Secondary resistance referred to primary side is given as

$$R_2' = \frac{R_2}{K^2}$$

The equivalent resistance referred to primary side is given as

$$R_{ep} = R_1 + R_2'$$

Secondary reactance referred to primary side is given as

$$X_2' = \frac{X_2}{K^2}$$

The equivalent reactance referred to primary side is given as

$$X_{ep} = X_1 + X_2'$$

Equivalent Circuit when all the Quantities are referred to Secondary side:

The equivalent circuit diagram of the transformer is shown below when all the quantities are referred to the secondary side.

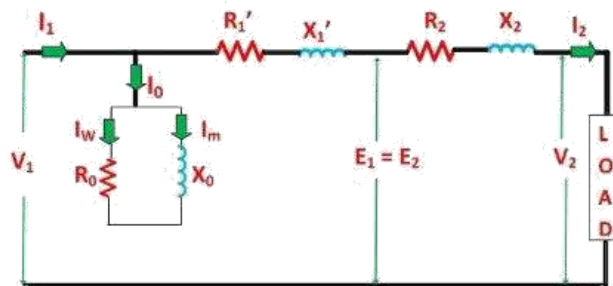


Fig: Circuit Diagram of Transformer When All the Primary Quantities are Referred to Secondary Side

The following are the values of resistance and reactance given below
Primary resistance referred to secondary side is given as

$$R_1' = K^2 R_1$$

The equivalent resistance referred to secondary side is given as

$$R_{es} = R_2 + R'_1$$

Primary reactance referred to secondary side is given as

$$X'_1 = K^2 X_1$$

The equivalent reactance referred to secondary side is given as

$$X_{eq} = X_2 + X'_1$$

No load current I_0 is hardly 3 to 5% of full load rated current, the parallel branch consisting of resistance R_0 and reactance X_0 can be omitted without introducing any appreciable error in the behavior of the transformer under the loaded condition.

Further simplification of the equivalent circuit of the transformer can be done by neglecting the parallel branch consisting R_0 and X_0 . The simplified circuit diagram of the transformer is shown below

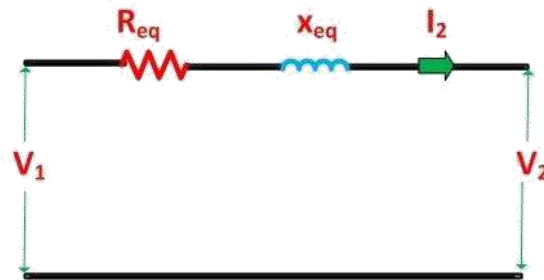


Fig: Simplified Equivalent Circuit Diagram of a Transformer

VOLTAGE REGULATION OF A TRANSFORMER:

Definition: The voltage regulation is defined as the change in the magnitude of receiving and sending the voltage of the transformer. The voltage regulation determines the ability of the transformer to provide the constant voltage for variable loads.

When the transformer is loaded with continuous supply voltage, the terminal voltage of the transformer varies. The variation of voltage depends on the load and its power factor. Mathematically, the voltage regulation is represented as

$$\text{Voltage Regulation} = \frac{E_2 - V_2}{E_2}$$

$$\% \text{ Voltage Regulation} = \frac{E_2 - V_2}{E_2} \times 100$$

Where,

E_2 – secondary terminal voltage at no load

V_2 – secondary terminal voltage at full load

The voltage regulation by considering the primary terminal voltage of the transformer is expressed as,

$$\% \text{ Voltage Regulation} = \frac{V_1 - E_1}{V_1} \times 100$$

Let us understand the voltage regulation by taking an example explained below

If the secondary terminals of the transformer are open circuited or no load is connected to the secondary terminals, the no-load current flows through it. If the no current flows through the secondary terminals of the transformer, the voltage drops across their resistive and reactive load become zero. The voltage drop across the primary side of the transformer is negligible.

If the transformer is fully loaded, i.e., the load is connected to their secondary terminal, the voltage drops appear across it. The value of the voltage regulation should always be less for the better performance of transformer.

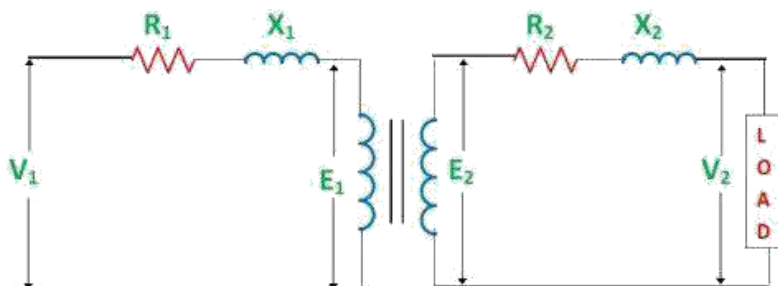


Fig: Equivalent Circuit of transformer

From the circuit diagram shown above, the following conclusions are made

- The primary voltage of the transformer is always greater than the emf induces on the primary side. $V_1 > E_1$
- The secondary terminal voltage at no load is always greater than the voltage at full load condition. $E_2 > V_2$

By considering the above circuit diagram, the following equations are drawn

$$V_1 = I_1 R_1 \cos \phi_1 + I_1 X_1 \sin \phi_1 + E_1$$

$$E_2 = I_2 R_2 \cos \phi_2 + I_2 X_2 \sin \phi_2 + V_2$$

The approximate expression for the no-load secondary voltage for the different types of load is

For inductive load:

$$E_2 = I_2 R_{02} \cos \phi_2 + I_2 X_{02} \sin \phi_2 + V_2$$

OR

$$E_2 - V_2 = I_2 R_{02} \cos \phi_2 + I_2 X_{02} \sin \phi_2$$

OR

$$\frac{E_2 - V_2}{E_2} \times 100 = \frac{I_2 R_{02}}{E_2} \times 100 \cos \phi_2 + \frac{I_2 X_{02}}{E_2} \times 100 \sin \phi_2$$

Where,

$\frac{I_2 R_{02}}{E_2} \times 100$ is a percentage resistance drop

$\frac{I_2 X_{02}}{E_2} \times 100$ is a percentage reactance drop

For Capacitive load:

$$E_2 = I_2 R_{02} \cos \phi_2 - I_2 X_{02} \sin \phi_2 + V_2$$

OR

$$E_2 - V_2 = I_2 R_{02} \cos \phi_2 - I_2 X_{02} \sin \phi_2$$

OR

$$\frac{E_2 - V_2}{E_2} \times 100 = \frac{I_2 R_{02}}{E_2} \times 100 \cos \phi_2 - \frac{I_2 X_{02}}{E_2} \times 100 \sin \phi_2$$

ALL DAY EFFICIENCY OF A TRANSFORMER:

Definition: All day efficiency means the power consumed by the transformer throughout the day. It is defined as the ratio of output power to the input power in kWh or Wh of the transformer over 24 hours.

Mathematically, it is represented as

$$\text{All day efficiency, } \eta_{\text{all day}} = \frac{\text{output in kWh}}{\text{input in kWh}} \quad (\text{for 24 hours})$$

All day efficiency of the transformer depends on their load cycle. The load cycle of the transformer means the repetitions of load on it for a specific period.

The ordinary or commercial efficiency of a transformer define as the ratio of the output power to the input power.

$$\eta = \frac{\text{output power}}{\text{input power}} = \frac{\text{output power}}{\text{output power} + \text{losses}}$$

What is the need of All Day Efficiency?

Some transformer efficiency cannot be judged by simple commercial efficiency as the load on certain transformer fluctuates throughout the day. For example, the distribution transformers are energised for 24 hours, but they deliver very light loads for the major portion of the day, and they do not supply rated or full load, and most of the time the distribution transformer has 50 to 75% load on it.

As we know, there are various losses in the transformer such as iron and copper loss. The iron loss takes place in the core of the transformer. Thus, the iron or core loss occurs for the whole day in the distribution transformer. The second type of loss known as copper loss takes place in the windings of the transformer also known as the variable loss. It occurs only when the transformers are in the loaded condition.

Hence, the performance of such transformers cannot be judged by the commercial or ordinary efficiency, but the efficiency is calculated or judged by All Day Efficiency also known as operational efficiency or energy efficiency which is computed by energy consumed during 24 hours.

POLARITY TEST OF TRANSFORMER:

Polarity means the direction of the induced voltages in the primary and the secondary winding of the transformer. If the two transformers are connected in parallel, then the polarity should be known for the